**Institut Teknologi Bandung**

Directorate of Continuing Professional Education

in partnership with

CHAPTER 8

# Image Classification

ConvNets from first principles, then the three strategies that make small image datasets tractable — from 80% to 98.6% on the same 2,000 pictures.

**Duration** 3 hours (2 sessions)

**Instructor** Rahman Indra Kesuma, S.Kom., M.Cs.

**Source** Chollet & Watson, Deep Learning with Python 3e — chapter 8

LEARNING OUTCOMES

# Session Objectives

By the end of this session, participants can:

1. Explain why a convolution layer beats a dense layer on images: **translation invariance** and **spatial hierarchies**.
2. Read a ConvNet summary and predict how **feature-map shape and depth** change layer by layer.
3. Say what **max pooling** is for, and what breaks in a ConvNet without it.
4. Build an image pipeline with `image_dataset_from_directory` and `tf.data`, including prefetching.
5. Apply **data augmentation** and explain what it can and cannot fix.
6. Use a pretrained model two ways — **fast feature extraction** and **end-to-end with augmentation** — and know the trade-off.
7. **Fine-tune** a pretrained base in the right order, at the right learning rate, and know which layers to leave frozen.

BOOK
Chapter 8 — full text

NOTEBOOK
01_convnet_on_mnist.ipynb

NOTEBOOK
02_small_dataset_from_scratch.ipynb

NOTEBOOK
03_data_augmentation.ipynb

NOTEBOOK
04_feature_extraction_and_finetuning.ipynb

REPO
Author's official notebook

ROADMAP

# One problem, three strategies, four numbers

**≈80%**
small ConvNet from scratch, no regularisation

**≈84%**
the same, with data augmentation

**98.1%**
feature extraction from a pretrained model

**98.6%**
fine-tuning that pretrained model

All four use the **same 2,000 training images**. The lesson of the chapter is in the gap between the second number and the third — **reusing learned features is worth more than any amount of tuning your own small model**

# 01

# Introduction to ConvNets

Why a convolution beats a dense layer on images.

# A basic ConvNet is a stack of two layer types

listing 8.1 — a small ConvNet PYTHON

```python
import keras
from keras import layers

inputs = keras.Input(shape=(28, 28, 1))
x = layers.Conv2D(filters=64, kernel_size=3, activation="relu")(inputs)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=128, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
x = layers.GlobalAveragePooling2D()(x)
outputs = layers.Dense(10, activation="softmax")(x)
model = keras.Model(inputs=inputs, outputs=outputs)
```

A ConvNet takes tensors of shape `(image_height, image_width, image_channels)`, not counting the batch dimension.

# What the summary tells you

OUTPUT

```
Layer (type)                 Output Shape              Param #
input_layer (InputLayer)     (None, 28, 28, 1)           0
conv2d (Conv2D)              (None, 26, 26, 64)        640
max_pooling2d (MaxPooling2D) (None, 13, 13, 64)          0
conv2d_1 (Conv2D)            (None, 11, 11, 128)    73,856
max_pooling2d_1 (MaxPooling2D)(None, 5, 5, 128)          0
conv2d_2 (Conv2D)            (None, 3, 3, 256)     295,168
global_average_pooling2d     (None, 256)                 0
dense (Dense)                (None, 10)              2,570

Total params: 372,234 (1.42 MB)
```

- **Width and height shrink** as you go deeper: 28 → 26 → 13 → 11 → 5 → 3.
- **Depth grows**: 1 → 64 → 128 → 256, controlled by the first argument to `Conv2D`.

# The result, against the dense network from chapter 2

listing 8.3 — training it PYTHON

```python
train_images = train_images.reshape((60000, 28, 28, 1)).astype("float32") / 255
test_images = test_images.reshape((10000, 28, 28, 1)).astype("float32") / 255

model.compile(optimizer="adam", loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])
model.fit(train_images, train_labels, epochs=5, batch_size=64)

test_loss, test_acc = model.evaluate(test_images, test_labels)
print(f"Test accuracy: {test_acc:.3f}")
```

OUTPUT

```
Test accuracy: 0.991
```

**97.8% → 99.1%**
dense network vs ConvNet

**≈ 60%**
relative reduction in error rate

SECTION 8.1.1

# The fundamental difference: global versus local


The one architectural change that produced that 60% error reduction.

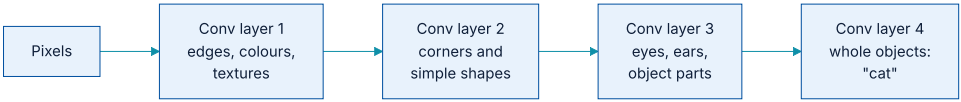
SECTION 8.1.1

# Property 1 — translation invariance

This makes ConvNets **data efficient** on images, because the visual world is fundamentally translation invariant. They **need fewer training samples** to learn representations that generalise.

SECTION 8.1.1

# Property 2 — spatial hierarchies




Elementary lines and textures combine into eyes and ears, which combine into high-level concepts.

This works because **the visual world is itself spatially hierarchical**. The architecture matches the structure of the data — which is exactly what chapter 5 called a good **architecture prior**

SECTION 8.1.1

# Feature maps, filters, and response maps

- The output's depth is **a parameter of the layer**, and its channels no longer stand for colours — they stand for **filters**.
- A filter encodes a specific aspect of the input; at a high level, one filter might encode *"presence of a face"*.
- In the MNIST example the first layer turns `(28, 28, 1)` into `(26, 26, 64)` — it computes **64 filters** over its input.
- Each of those 64 channels is a $26 \times 26$ grid: a **response map** saying **how strongly that filter's pattern appears at each location**.

SECTION 8.1.1

# How the convolution actually runs

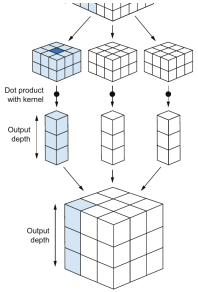



Figure 8.4 — a window slides over the input, each 3D patch is turned into a vector by the same learned kernel, and the vectors are reassembled into an output map. — Chollet & Watson, *Deep Learning with Python*, 3rd ed. (Manning)

SECTION 8.1.1

# The two parameters that define a convolution

| PARAMETER | TYPICAL VALUE | WHAT IT CONTROLS |
|---|---|---|
| **Patch size** | $3 \times 3$ or $5 \times 5$ | How large a neighbourhood each output value can see. |
| **Output depth** | 32 to 512 | How many filters are computed — how many different patterns the layer can look for. |

how Keras spells it — PYTHON

```python
layers.Conv2D(output_depth, (window_height, window_width))

# the same kernel is reused for every patch, which is why the layer is
# translation invariant and why it has so few parameters
```

Because the kernel is shared across all positions, **parameter count depends on the window and depth, not on the image size**

SECTION 8.1.2

# Max pooling: downsample, aggressively

| | CONVOLUTION | MAX POOLING |
|---|---|---|
| Transformation | A **learned** linear transform (the kernel). | A **hardcoded** max operation. |
| Typical window | $3 \times 3$, stride 1 | $2 \times 2$, stride 2 |
| Effect on size | Shrinks slightly (no padding). | **Halves** each spatial dimension. |

# What happens without it

listing 8.5 — an incorrectly structured ConvNet PYTHON

```python
inputs = keras.Input(shape=(28, 28, 1))
x = layers.Conv2D(filters=64, kernel_size=3, activation="relu")(inputs)
x = layers.Conv2D(filters=128, kernel_size=3, activation="relu")(x)
x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
x = layers.GlobalAveragePooling2D()(x)
outputs = layers.Dense(10, activation="softmax")(x)
model_no_max_pool = keras.Model(inputs=inputs, outputs=outputs)
```

**It is not conducive to learning a spatial hierarchy.** The $3 \times 3$ windows in the third layer still only see information from **$7 \times 7$ windows of the original image** — far too small a view to assemble a whole digit.

# 02

# Training from scratch on a small dataset

2,000 pictures of cats and dogs. A situation you will actually meet.

SECTION 8.2

# The setup, and why it is realistic

**2,000**
training images

**1,000**
validation images

**2,000**
test images

Drawn from the original Kaggle Dogs vs. Cats dataset, which was released as a competition in late 2013 — **before ConvNets were mainstream**

SECTION 8.2.1

# Why deep learning still applies with little data

**ConvNets are data efficient**

Because they learn **local, translation-invariant** features, they give reasonable results on small image datasets without any custom feature engineering.

**Models are repurposable**

A classifier trained on a large dataset can be reused on a significantly different problem with minor changes. **Feature reuse is one of deep learning's greatest strengths.**

What counts as *enough* is relative — to the size and depth of the model. A few tens of samples will not do; a few hundred can suffice if the model is small, well regularised, and the task is simple.

# A bigger ConvNet for bigger images

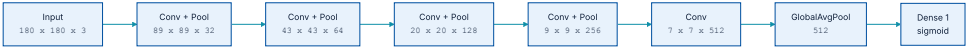
listing 8.7 — dogs vs cats, from scratch PYTHON

```python
inputs = keras.Input(shape=(180, 180, 3))
x = layers.Rescaling(1.0 / 255)(inputs)          # 0..255 -> 0..1, inside the model
x = layers.Conv2D(filters=32, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=64, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=128, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=512, kernel_size=3, activation="relu")(x)
x = layers.GlobalAveragePooling2D()(x)
outputs = layers.Dense(1, activation="sigmoid")(x)     # binary -> one sigmoid unit
model = keras.Model(inputs=inputs, outputs=outputs)
```

Putting `Rescaling` in the model means the deployed artefact carries its own preprocessing — one fewer thing to get wrong in production.

SECTION 8.2.3

# The pattern you will see in almost every ConvNet




Depth rises from 32 to 512 while the spatial size falls from $180 \times 180$ to $7 \times 7$.

**Feature-map depth increases while feature-map size decreases.** Information moves out of *where* and into *what* — **and that is the whole shape of a classification ConvNet**

SECTION 8.2.4

# From JPEG files to batched tensors

1. Read the picture files.
2. Decode the JPEG content into RGB grids of pixels.
3. Convert those into floating-point tensors.
4. Resize them to a shared size — here $180 \times 180$.
5. Pack them into batches.

SECTION 8.2.4 · LISTING 8.9

# `image_dataset_from_directory` does all five

listing 8.9 — building the pipeline PYTHON

```python
from keras.utils import image_dataset_from_directory

batch_size = 64
image_size = (180, 180)

train_dataset = image_dataset_from_directory(
    new_base_dir / "train", image_size=image_size, batch_size=batch_size)
validation_dataset = image_dataset_from_directory(
    new_base_dir / "validation", image_size=image_size, batch_size=batch_size)
test_dataset = image_dataset_from_directory(
    new_base_dir / "test", image_size=image_size, batch_size=batch_size)
```

The directory layout **is** the label set: one subfolder per class, and the folder names become the classes.

SECTION 8.2.4

# What a `tf.data.Dataset` buys you

- It works with **any backend** — JAX and PyTorch included — not only TensorFlow.
- It is an **iterator**: usable in a `for` loop, and passable straight to `fit()`.
- It **parallelises preprocessing across CPU cores**.
- It does **asynchronous prefetching**: preparing the next batch while the model is still working on the previous one, so **execution never stalls**.

> Those last two are the difference between a GPU that is busy and a GPU that spends most of its time waiting for JPEGs to decode.

# 03

# Data augmentation

Making more training data out of the training data you have.

SECTION 8.2.5

# The idea, and where the layers go

| WHERE YOU PUT THE LAYERS | RUNS ON | TRADE-OFF |
|---|---|---|
| **Inside the model**, before `Rescaling` | GPU | Sometimes faster, but the augmentation ships with the model. |
| **In the data pipeline**, via `map()` | CPU, in parallel | **Usually the better option** — and the one the book takes. |

# Three transformations, and prefetching

listing 8.18 — the augmentation stage PYTHON

```python
data_augmentation_layers = [
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.1),
    layers.RandomZoom(0.2),
]

def data_augmentation(images, targets):
    for layer in data_augmentation_layers:
        images = layer(images)
    return images, targets

augmented_train_dataset = train_dataset.map(data_augmentation, num_parallel_calls=8)
augmented_train_dataset = augmented_train_dataset.prefetch(tf.data.AUTOTUNE)
```

- `RandomFlip("horizontal")` — flips a random **50%** of images.
- `RandomRotation(0.1)` — rotates by a random value in **[−10%, +10%]** of a full circle, i.e. ±**36 degrees**.
- `RandomZoom(0.2)` — zooms in or out by a random factor in **[−20%, +20%]**.

SECTION 8.2.5

# What augmentation cannot do

The model will never see the same input twice — but the inputs it does see are **still heavily intercorrelated**, because they come from a small number of originals. **You cannot produce new information; you can only remix existing information.**

**≈ 80%**
from scratch, no augmentation

**≈ 84%**
from scratch, with augmentation

Four points for a change that costs nothing but CPU time — worth having, and clearly not enough on its own.

# 04

# Using a pretrained model

Where the remaining fourteen points come from.

SECTION 8.3

# Why features transfer between problems

The book uses **Xception** trained on ImageNet: 1.4 million labelled images across 1,000 classes. ImageNet contains many cat and dog breeds, so it should transfer well here.

> Such portability across problems is **a key advantage of deep learning over older shallow methods**, and it is what makes it effective on small-data problems at all.

SECTION 8.3

# Two ways to reuse it


Feature extraction swaps the classifier; fine-tuning then also adjusts the top of the base.

SECTION 8.3.1

# Why only the convolutional base is reused

- The base produces **presence maps of generic concepts** — useful regardless of the vision problem.
- The classifier's representations are **specific to the classes it was trained on**.
- Worse, dense layers **discard spatial information**: they no longer know *where* anything is. For problems where location matters they are **largely useless**.

SECTION 8.3.1

# How much of the base to reuse

So if your new dataset differs a lot from the original, you may be better off using **only the first few layers** for feature extraction rather than the entire base.

# Loading a pretrained backbone

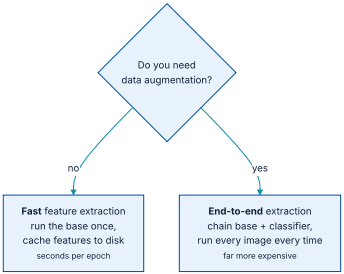
listing 8.23–8.24 — backbone and preprocessing PYTHON

```python
import keras_hub

conv_base = keras_hub.models.Backbone.from_preset("xception_41_imagenet")

preprocessor = keras_hub.layers.ImageConverter.from_preset(
    "xception_41_imagenet",
    image_size=(180, 180),
)
```

- **backbone** is KerasHub's word for the feature extractor without its classification head.
- The **41** is a naming convention: 41 trainable layers. Pretrained ConvNets are conventionally named by depth.
- `ImageConverter` matters more than it looks: every pretrained ConvNet expects a particular input range, and feeding the wrong one forces the model to **relearn how to see**.

SECTION 8.3.1

# Fast, or augmentable — pick one




The base is by far the most expensive part of the pipeline, so running it once versus every epoch is the whole trade-off.

SECTION 8.3.1 · LISTING 8.25

# Fast route — cache the features once

listing 8.25 — extracting features PYTHON

```python
def get_features_and_labels(dataset):
    all_features, all_labels = [], []
    for images, labels in dataset:
        preprocessed_images = preprocessor(images)
        features = conv_base.predict(preprocessed_images, verbose=0)
        all_features.append(features)
        all_labels.append(labels)
    return np.concatenate(all_features), np.concatenate(all_labels)

train_features, train_labels = get_features_and_labels(train_dataset)
val_features, val_labels = get_features_and_labels(validation_dataset)
test_features, test_labels = get_features_and_labels(test_dataset)

print(train_features.shape)
```

OUTPUT

```
(2000, 6, 6, 2048)
```

# …then train a tiny classifier on the cache

listing 8.26 — the dense classifier PYTHON

```python
inputs = keras.Input(shape=(6, 6, 2048))
x = layers.GlobalAveragePooling2D()(inputs)
x = layers.Dense(256, activation="relu")(x)
x = layers.Dropout(0.25)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model = keras.Model(inputs, outputs)
model.compile(loss="binary_crossentropy", optimizer="adam", metrics=["accuracy"])

history = model.fit(train_features, train_labels, epochs=10,
                    validation_data=(val_features, val_labels),
                    callbacks=callbacks)
```

OUTPUT

```
validation accuracy: slightly over 98%
test accuracy      : 0.981
(an epoch takes under 1 second, even on CPU)
```

SECTION 8.3.1

# Read that result carefully

> It is **a slightly unfair comparison**: ImageNet contains many dog and cat instances, so the pretrained model already holds **the exact knowledge this task needs**. That will not always be true.

The curves also show **overfitting almost from the start**, despite a fairly large dropout rate — because this route cannot use augmentation, which is essential on small image datasets.

# Slow route — freeze the base and chain it

listing 8.28–8.29 — freezing PYTHON

```python
conv_base = keras_hub.models.Backbone.from_preset(
    "xception_41_imagenet",
    trainable=False,
)

conv_base.trainable = True
print(len(conv_base.trainable_weights))     # before freezing
conv_base.trainable = False
print(len(conv_base.trainable_weights))     # after freezing
```

OUTPUT

```
26
0
```

Setting `trainable = False` **empties the list of trainable weights** — which is exactly how you verify that a freeze took effect.

SECTION 8.3.1

# The chained model, and a recompile trap

the end-to-end model

PYTHON

```python
inputs = keras.Input(shape=(180, 180, 3))
x = preprocessor(inputs)
x = conv_base(x)
x = layers.GlobalAveragePooling2D()(x)
x = layers.Dense(256)(x)
x = layers.Dropout(0.25)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model = keras.Model(inputs, outputs)
model.compile(loss="binary_crossentropy", optimizer="adam", metrics=["accuracy"])
```

Only the two `Dense` layers train — four weight tensors in total. And a trap worth memorising: **if you change trainability after compiling, you must recompile**, or **the change is silently ignored**

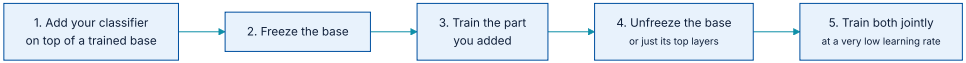
# 05

# Fine-tuning

Unfreezing the top of the base — carefully, and last.

SECTION 8.3.2

# The five steps, and why the order is fixed




Steps 1–3 are exactly the feature-extraction workflow. Fine-tuning is what you do afterwards.

The base can only be fine-tuned **once the classifier on top is already trained**. Otherwise the error signal propagating back is too large, and **the representations being fine-tuned are destroyed**

SECTION 8.3.2

# Partial fine-tuning, and why the top layers

- Earlier layers encode **generic, reusable** features; higher ones encode **specialised** features.
- It is the specialised ones that need repurposing, so there are **fast-decreasing returns** to fine-tuning lower layers.
- The base has **15 million parameters** — training all of them on 2,000 images risks overfitting.

unfreezing the top four — PYTHON

```python
conv_base.trainable = True
for layer in conv_base.layers[:-4]:
    layer.trainable = False
```

One layer type to leave alone: **do not unfreeze `BatchNormalization` layers**. Chapter 9 explains why.

# Fine-tune at a very low learning rate

listing 8.31 — fine-tuning PYTHON

```python
model.compile(
    loss="binary_crossentropy",
    optimizer=keras.optimizers.Adam(learning_rate=1e-5),    # note: 1e-5
    metrics=["accuracy"],
)

callbacks = [keras.callbacks.ModelCheckpoint(
    filepath="fine_tuning.keras", save_best_only=True, monitor="val_loss")]

history = model.fit(augmented_train_dataset, epochs=30,
                    validation_data=validation_dataset, callbacks=callbacks)
```

> Updates that are too large **harm the representations you are trying to keep**. This is one of the few places in the book where the learning rate is prescribed rather than searched for.

SECTION 8.3.2

# The final number, in context

evaluating the fine-tuned model — PYTHON

```python
model = keras.models.load_model("fine_tuning.keras")
test_loss, test_acc = model.evaluate(test_dataset)
print(f"Test accuracy: {test_acc:.3f}")
```

OUTPUT

```
Test accuracy: 0.986
```

In the original Kaggle competition this **would have been among the top results** — though not a fair comparison, since the pretrained features already carried prior knowledge about cats and dogs that competitors could not use.

> The fairer point: this was reached using **about 10% of the training data** available in that competition. **There is a huge difference between training on 20,000 samples and on 2,000.**

SUMMARY

# What has to survive this chapter

1. **Convolutions learn local patterns**, which buys translation invariance and spatial hierarchies — and makes ConvNets data efficient on images.
2. **Depth rises as spatial size falls.** That shape is nearly universal in classification ConvNets.
3. **Max pooling is not optional**: without it, deep layers still only see a tiny window of the original image.
4. `image_dataset_from_directory` + `tf.data` turns a folder of JPEGs into a parallel, prefetching pipeline.
5. **Augmentation remixes information; it cannot create it.** Worth about four points here.

SUMMARY

# . . . and the part that changes how you plan a project

**Reuse beats tuning**

Feature extraction from a pretrained base took the same 2,000 images from **84% to 98.1%** — more than anything you could do to a small model of your own.

**Fine-tune last, and gently**

Only after the new classifier is trained, at a **very low learning rate**, and with `BatchNormalization` left frozen.

Which is why the first question on a new vision problem is not *what architecture?* but ***whose features can I start from?***

NOTEBOOK

04_feature_extraction_and_finetuning.ipynb

NEXT

Chapter 9 — ConvNet architecture patterns